



Data Analytics for Business Applications

Analysis Report

Objective

The analysis focuses on understanding user engagement with audiobooks using different machine learning techniques: Correlation Analysis, Linear Regression, Random Forest, and K-Means Clustering.

Key Findings & Insights

1. Correlation Analysis & Linear Regression

- Features analyzed: **Ratings_Given**, **Reviews_Written**, **Social_Sharing** as predictors for **Number_of_Audiobooks_Completed**.
- The Linear Regression model produced a Mean Squared Error (MSE), indicating moderate predictive power.
- The coefficients suggest a positive correlation between user engagement (ratings, reviews, social sharing) and audiobook completion.

2. Random Forest Model

- Applied as an alternative to Linear Regression for better feature importance insights.
- Root Mean Squared Error (RMSE) was lower than Linear Regression, indicating better performance.
- Feature Importance:
 - **Ratings_Given** was the most influential factor in predicting **Number_of_Audiobooks_Completed**.

- **Social_Sharing** and **Reviews_Written** had lesser importance but still contributed.

3. K-Means Clustering

- Segmented users into 3 distinct clusters based on:
 - **Genre_Exploration**
 - **Number_of_Audiobooks_Completed**
 - **Membership_Duration**
- Cluster Characteristics:
 - Cluster 1: Low engagement users with short membership duration.
 - Cluster 2: Moderate engagement users exploring various genres.
 - Cluster 3: Highly engaged users with long membership duration and high completion rates.
- Subscription type, age group, and gender showed some cluster-based patterns.

Recommendations

1. User Engagement Strategies

- Increase **Ratings_Given** via gamification (badges, leaderboards).
- Incentivize **Social_Sharing** with referral programs.

2. Targeted Marketing

- Focus promotions on high-engagement users to boost audiobook consumption.
- Personalize recommendations for low-engagement users to improve retention.

3. Product Enhancements

- Introduce features to encourage **Reviews_Written**, as it correlates with engagement.
- Improve subscription models tailored to different clusters.

Conclusion

This analysis highlights the factors influencing audiobook completion and provides actionable insights to enhance user engagement through targeted strategies.